CTX700 K-Line Dash

A DIY Bluetooth diagnostic dashboard for the 2016 Honda CTX700 (NC700 platform)

Collaboration & development brief | License: Apache-2.0 | Compiled 2026-11-05

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1. Overview

This project reads live engine data from a 2016 Honda CTX700 over its factory K-line diagnostic bus and streams it to a phone over Bluetooth LE. The CTX700 does not speak OBD-II; it uses Honda's proprietary K-line protocol (a KWP2000 variant that is not standard ISO-9141), so generic ELM327 dongles cannot read it. The design is headless: a thumbnail-sized microcontroller plus a K-line transceiver seal at the diagnostic connector and broadcast to a phone, which acts as the display. No wiring runs to the dash.

Goals: cheap, self-contained, read-only, phone-as-dash, open for others to extend.

2. System architecture

```
Bike HM090 4-pin ---- K-line -------> L9637D  K pin
   (K / +12V / GND)   +12V (sw) ----> fuse --> LM2596 buck --> 5V --> XIAO
                                      +12V ---> 510 ohm pull-up to K-line
                                      +12V ---> 1N4007 --> L9637D Vbatt
                      Ground -------> common ground (bike + L9637D + XIAO)

L9637D  TX-in  <--- XIAO D6 (TX)      L9637D RX-out ---> XIAO D7 (RX)

XIAO  ))) Bluetooth LE ))) phone  (Web Bluetooth dashboard / BLE terminal)
```

The K-line is a single half-duplex wire; the transceiver echoes transmitted bytes back, so the firmware discards its own echo before reading each ECU reply.

3. Hardware & bill of materials

Part	Role	Notes
XIAO ESP32-S3	MCU + BLE	BLE-only; S3 is dual-core. C3/C6 work (change UART pins).
L9637D (SO-8)	K-line transceiver	ISO-9141 physical layer. The enabling part.
510 ohm resistor	K-line pull-up to +12V	Idle-high bias. 470-560 ohm also fine.
1N4007 diode	Reverse-polarity protection	On the +12V feed; stripe (cathode) toward the chip.
LM2596 buck	12V -> 5V regulator	Set to 5.0 V before connecting.
Sumitomo HM090 4-pin	Bike diagnostic plug	Pins: K-line / +12V (ign) / GND / reset (unused).
Inline fuse 1-2 A	Protect bike harness	On the +12V tap.
3D-printed pod	Sealed enclosure	Potted; keep the XIAO antenna end above the pour.

UART pins (XIAO labels D6=TX, D7=RX): S3 = GPIO43/44, C3 = GPIO21/20, C6 = GPIO16/17.

4. K-line protocol (verified from documentation)

Link: K-line, **10400 baud, 8-N-1**. Not standard ISO-9141 (why ELM327 fails). Sourced from the gonzos.net CTX-OBD reverse-engineering and cross-checked against the andreibaw and roccofz1 Honda K-line projects and the Honda data tables.

4.1 Init handshake

```
1. Drive K-line LOW 70 ms, then HIGH 120 ms
2. Send wake-up:  FE 04 FF FF   then wait 200 ms
3. Send init:     72 05 00 F0 99
4. ECU replies:   02 04 00 FA
```

4.2 Checksum (every message)

```
checksum = 0x100 - (sum of all preceding bytes), masked to 8 bits
e.g.  72 05 71 10  ->  sum 0xF8  ->  checksum 0x08
```

4.3 Table request / response

```
request:   72 05 71 [table] [checksum]
response:  02 [len] 71 [table] [data ...] [checksum]
           data begins at response index 4 (after the 4-byte header)
```

CTX700 uses sensor table **0x11** (and 0x61); try 0x10 if 0x11 is silent. Gear/stand status is in a separate table **0xD1**.

4.4 Sensor byte map (table 0x11, byte # within the data payload)

Byte	Parameter	Convert raw -> value
1-2	RPM	16-bit, = RPM
3	TPS voltage	raw x 5 / 256
4	Throttle %	throttle position
5	ECT voltage	raw x 5 / 256
6	Coolant temp C	raw - 40
7	IAT voltage	raw x 5 / 256
8	Intake air C	raw - 40
9	MAP voltage	raw x 5 / 256
10	MAP (kPa)	raw
11-12	(unused)	reads 0xFF
13	Battery V	raw / 10
14	Speed	raw (km/h)
15-16	Fuel injection	16-bit; multiplier UNKNOWN (see open problems)

4.5 Gear / stand status (table 0xD1, first byte)

```
0 = in gear    1 = neutral / clutch pulled    3 = kickstand down
```

This is a state, not a 1-6 gear number; the bus does not report selected gear.

5. Bluetooth LE layer

Transport: **Nordic UART Service (NUS)**, the de-facto BLE serial profile, via GATT notifications. Implemented with NimBLE-Arduino v2.x. Advertised as **CTX700-Dash**; clients should filter on the service UUID (the 128-bit UUID plus name can overflow the 31-byte advert, so name-only filtering is unreliable).

```
Service  (NUS):  6E400001-B5A3-F393-E0A9-E50E24DCCA9E
TX char  notify: 6E400003-B5A3-F393-E0A9-E50E24DCCA9E   (XIAO -> phone, data)
RX char  write:  6E400002-B5A3-F393-E0A9-E50E24DCCA9E   (phone -> XIAO, unused)
```

5.1 Data format

Newline-terminated ASCII CSV, prefixed **D,** ; raw SI units (client converts to F / mph). Non-**D,** lines are human-readable status (e.g. 'ECU linked') and are ignored by the app.

```
D,rpm,ectC,iatC,tps,mapkPa,batt,speedKmh,stand\r\n
example:  D,3200,92,34,18,61,14.1,74,0
```

6. Firmware (ctx700_kline_ble.ino)

Arduino / ESP32 core, NimBLE-Arduino v2.x. Read-only by design.

- Slow-init handshake, then polls tables 0x11 and 0xD1 at ~5 Hz.
- Half-duplex echo cancellation on the shared K wire; per-frame checksum verification (bad frames rejected).
- Reconnect: re-runs the ECU handshake on key-on; resumes BLE advertising on phone disconnect.
- DEMO flag (one const): streams synthetic moving data over BLE with no bike/transceiver, for validating the app end.
- Emits compact CSV over BLE and a human-readable line over USB serial for bench debugging.

7. Phone dashboard (ctx700_dash.html)

Single self-contained HTML page using Web Bluetooth. Connects to the NUS service, subscribes to the TX characteristic, parses the CSV, and renders gauges (RPM bar with redline, coolant with warning colors, speed, throttle, battery, MAP, neutral/gear/kickstand). Auto-reconnects.

- Requires Chrome on Android, served over HTTPS (e.g. GitHub Pages). iOS Safari has no Web Bluetooth.
- Filters on the NUS service UUID for reliable discovery.
- Alternative clients: Serial Bluetooth Terminal (BLE mode), nRF Connect.

8. Status & open problems (collaboration asks)

Validated:

- Firmware compiles and runs on XIAO ESP32-S3; advertises over BLE.
- Phone connects via Web Bluetooth; dashboard renders; DEMO data drives all gauges.

Not yet validated on a live CTX700 ECU:

- The init handshake and byte map against the actual bike (documented, not yet bench-confirmed on hardware).
- L9637D logic sense (the K-line pulse polarity may need inverting for a given transceiver wiring).
- Whether the CTX700 answers on table 0x11 or 0x10.

Open questions worth collaborator input:

- Fuel-injection multiplier (bytes 15-16) is undocumented across every public Honda K-line project; needs empirical calibration against known fuel flow or tank mileage.
- A true 1-6 gear readout is not on the bus; it would require inferring gear from the speed/RPM ratio.
- Confirming the NC700-platform byte map holds exactly for the CTX700 vs. shared CBR600RR (PC41) tables.

9. References

- gonzos.net CTX-OBD project - https://gonzos.net/projects/ctx-obd/ (protocol + HM090 connector)
- andreibaw/Honda_K-Line_KWP2000 - K-line decoder, bench-tested on '05 CBR600RR
- roccofz1/HondaECU - Honda ECU K-line reverse engineering
- sophienyaa/Honda-Motorcycle-Dash - ESP32 + ILI9341 Honda dash (original display inspiration)
- muki01/OBD2_K-line_Reader - K-line transceiver reference circuits
- Honda data tables (tables 11/61) - cross-bike parameter map
- ST L9637D datasheet - K-line transceiver
- NimBLE-Arduino - BLE stack

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